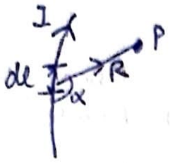
UNIT-2
Magnetostatics

Magnetostatics is the study of magnetic fields in systems where the currents are steady (not changing with time)

Biot Savart's Law (ET2017, 2019, and explain B & H)(5M)

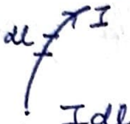


The magnetic field intensity dH produced at a point P by the differential current element Idl is

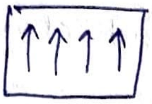
$$\boxed{\vec{H} = \frac{\mu_0}{4\pi\mu}\int_L \frac{I\vec{dl} \times \vec{R}}{R^3}} \quad A/m$$

$$\vec{dH} = \frac{K\,I\,dl\,\sin\alpha}{R^2}$$

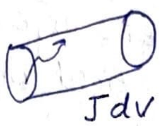
Current Distribution



Idl
Line current density
unit I → A



Kds
Surface charge density
K = A/m



Jdv
Volume charge density
J = A/m²

$Idl \equiv Kds \equiv Jdv$ Identically equal

Right Hand Rule :- Point thumb in direction of current then curl of fingers will point in direction of $\vec{H}$.

Magnetic Induction

The magnetism acquired by a magnetic material when it is kept near (or in contact) with a magnet. This magnetism is temperory.

→ The chain formation of chips due to induced magnetism will not be ∞. The chips will drop as soon as the weight of chips > force of attraction.

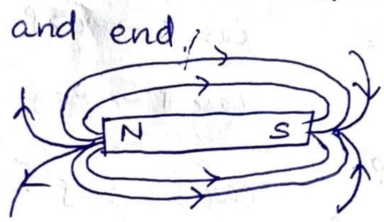
Magnetic Field Strength (H) and Magnetic Flux Density (B)

Magnetic flux line is a path to which B is tangential at every point on the line. It is the line along which the needle of a magnetic compass will orient itself if placed in the presence of a magnetic field.

Magnetic flux lines due to a straight wire with current coming out of page

→ The flux lines are closed and have no beginning and end,

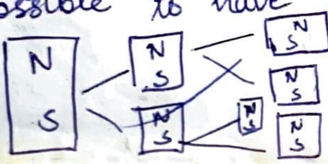
→



Magnetic flux density (B) and magnetic field intensity (H) are related as

$\vec{B} = \mu_0 \vec{H}$ where $\mu_0 = 4\pi \times 10^{-7}$ H/m → Permeability of free space.

Magnetic field lines always close upon themselves because it is not possible to have isolated magnetic poles



Successive division of a bar magnet results in pieces with north and south pole, showing that magnetic poles can't be isolated.

② The total flux passing through any closed surface in a magnetic field must be zero i.e.

$$\boxed{\oint \vec{B}\cdot\vec{ds} = 0} \longrightarrow \text{Maxwell's equation}$$

This is referred to as law of conservation of magnetic flux or Gauss law for magnetostatic field. Although magnetostatic field is not conservative, magnetic flux is conserved.

By applying Divergence theorem

$$\oint \vec{B}\cdot\vec{ds} = \int_V \nabla\cdot B\, dv = 0$$

Que: State that magnetic field has no source or sink (2.5)(ET 2019)

$$\boxed{\nabla\cdot B = 0} \rightarrow \text{Non-existence of magnetic monopole}$$

Ampere's Circuital Law (ET 2018, ET 2019 (2.5M)

Line integral of magnetic field intensity around closed loop is equal to total current enclosed by closed loop.

→ This closed loop is called amperian path where
1. $\vec{H}$ is normal or tangential to path
2. $\vec{H}$ has same value at all points of path.

$$\oint \vec{H}\cdot\vec{dl} = I_{enclosed}$$

$I = \vec{J}\cdot\vec{ds}$ → where J is current density (not volume current density)

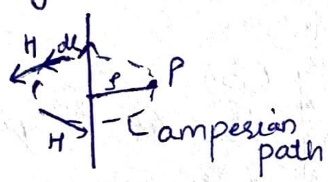
$$\boxed{\therefore \oint \vec{H}\cdot\vec{dl} = \int_S \vec{J}\cdot\vec{ds}}$$

Acc. to stroke's theorem → $\oint \vec{H}\cdot dl = \int (\nabla\times\vec{H})\, ds$

$$\therefore \oint(\nabla\times\vec{H})\, ds = \int_S \vec{J}\cdot d\vec{S}$$

$$\boxed{\nabla\times\vec{H} = J} \rightarrow \text{Maxwell's 3}^{rd}\text{ equation}$$

Magnetic Field Intensity due to current Element



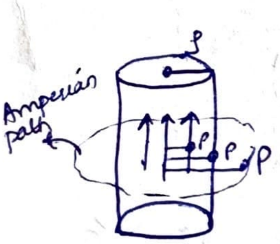
$$\oint \vec{H}\cdot\vec{dl} = I_{enclosed}$$

$$\oint H\cdot dl\cos\theta = I$$

$$H\oint dl = I$$

$$H = \frac{I}{2\pi\rho}\hat{a}_\phi$$

Magnetic Field Intensity due to conductor carrying current I



case I → $r > \rho$ $\quad \vec{H} = \frac{I}{2\pi r}\hat{a}_\phi$

case II → $r = \rho$ $\quad \vec{H} = \frac{I}{2\pi r}\hat{a}_\phi$ or $\frac{I}{2\pi\rho}\hat{a}_\phi$

case III $r < \rho$ ∴ $I_{enclosed} = \left(\frac{\text{Desired area}}{\text{Total area}}\right) \times I$

$$\oint \vec{H}\cdot dl = \left(\frac{\pi r^2}{\pi\rho^2}\right) I$$

$$H\cdot 2\pi r = \frac{\pi r^2 I}{\pi\rho^2} \rightarrow H = \frac{Ir}{2\pi\rho^2}\hat{a}_\phi$$

Que: In the region $0<\rho<0.5m$ in cylindrical coordinate, the current density is $J = 4.5 e^{-2\rho} \hat{a}_z$ A/m and $J=0$ elsewhere. Use Ampere Law to find H (ET 2019, 7.5)

(3)

Sol^n^:

$$\oint \vec{H}\cdot\vec{dl} = \int_S \vec{J}\cdot\vec{ds}$$

$$\vec{ds} = \rho\, d\rho\, d\phi\, \hat{a}_z$$

$$\int_S \vec{J}\cdot\vec{ds} = \int_S 4.5 e^{-2\rho} \hat{a}_z \rho\, d\rho\, d\phi\, \hat{a}_z = \int_{\phi=0}^{2\pi}\int_{\rho=0}^{0.5} 4.5 e^{-2\rho} d\rho\, d\phi = 4.5\times 2\pi \int_{\rho=0}^{0.5} \rho e^{-2\rho} d\rho = \cancel{4.5\times 2\pi\left[\frac{e^{-2\rho}}{-2}\right]_0^{0.5}}$$

$$\int_S \vec{J}\cdot\vec{ds} = 4.5\times 2\pi \times\left[\left(\frac{-\rho e^{-z\rho}}{z}\right)+\left(-\frac{e^{-\rho z}}{z^2}\right)\right] = -9\pi\left[\frac{1}{2z^2}\left(z e^{-z/2} + e^{-z/2}\right)\right.$$

$$\oint \vec{H}\cdot\vec{dl} = \oint H\cdot d\rho \cos 0^\circ = H\oint d\rho = H\cdot 2\pi\rho.$$

$dl = d\rho\, \vec{a}_\rho$

$$\oint \vec{H}\cdot\vec{dl} = \int_S \vec{J}\cdot\vec{ds}$$

$$H\cdot 2\pi\rho\, \hat{a}_\rho = \cancel{4.5}^{\,9}\times \cancel{2\pi}\times\left[\frac{(-e^{-z/2})}{2z} - \frac{e^{-z/2}}{z^2}\right]$$

$$\boxed{\vec{H} = \cancel{-\frac{4.5 e^{-z/2}}{2\rho}\hat{a}_\rho}}$$

$$\vec{H} = \frac{-9}{2z^2}\left[z e^{-z/2} - 2e^{-z/2}\right]\hat{a}_\rho$$

Faraday's law → follow law of conservation of energy.

1. Whenever there is a change in magnetic flux, emf will be induced and the emf lasts as long as the change in flux continues.
2. The emf induced is directly proportional to rate of change of flux.

Faraday's Law of Electromagnetic Induction (ET2018, 6M)

→ stationary magnetic field do not produce current.

Faraday & Henry discovered that a time varying magnetic field produce an electric current. Acc. to faraday's experiment a static magnetic field produce no current flow, but a time varying field produces an induced voltage (called electromotive force or simply emf) in a closed ckt, which causes flow of current.

Induced emf is equal to time rate of change of magnetic flux linkage

$$V_{emf} = -\frac{d\lambda}{dt} = -N\frac{d\psi}{dt} \quad \text{— (1)}$$

λ = Flux linkage
ψ = Flux through each turn
N = no. of turns

−ve sign indicate that the induced voltage opposes the flux produceing it. This is known as Lenz law.

Also, emf in a ckt is represented as line integral of electric field around a closed ckt or changing magnetic field that produces an electric field E_e

$$V = \oint \vec{E_e}\cdot\vec{dl} \quad \text{— (2)}$$

$\vec{E_e}$ is different from $\vec{E}$ due to charge
$\vec{E} = -\nabla V$ but $\vec{E_e} \neq -\nabla V$

$\vec{E_e}$ may be regarded as emf producing field.

We also know

$$\psi = \iint \vec{B}\cdot\vec{ds} \quad \text{— (3)}$$

Put (3) in (1) $\quad V = -\frac{d}{dt}\iint_S \vec{B}\cdot\vec{ds}$ ← This implies that a changing magnetic field will produce/induce an electric field

$$V = \oint \vec{E_e}\cdot\vec{dl} = -\frac{d}{dt}\iint_S \vec{B}\cdot\vec{ds}$$

or $V = \oint \vec{E_e}\cdot\vec{dl} = -\iint_S \frac{\partial\vec{B}}{\partial t}\cdot\vec{ds}$ → This is Faraday's Law of Induction

$\oint \vec{E_e}\cdot\vec{dl}$ is called induced electromotive force where $\vec{E_e}$ is the induced electric field intensity

Apply strokes theorem

$$V = \iint_S \nabla\times\vec{E_e}\, ds = -\iint_S \frac{\partial B}{\partial t}\cdot\vec{ds} \Rightarrow \nabla\times\vec{E_e} = -\frac{\partial B}{\partial t} \quad \text{or} \quad \boxed{\nabla\times\vec{E} = -\frac{\partial B}{\partial t}}$$

↓

Maxwell's equation for time varying magnetic field.

Que:- In cylindrical coordinates, $B = \frac{2}{r}\hat{a}_\phi$ Tesla. Find out the
(4) flux ϕ crossing the plane surface defined by $0.5 \le r \le 2.5m$ &
$0 \le z \le 2m$. (Sessional 2019)

Sol: $B = \frac{2}{r}\hat{a}_\phi$ Range: $0.5 \le r \le 2.5m$, $0 \le z \le 2m$ $ds = dr\,dz\,\hat{a}_\phi$

$$\psi = \int \vec{B}\cdot ds = \int_{z=0}^{2}\int_{r=\frac{1}{2}}^{r=2.5} \frac{2}{r}\,dr\,dz = 2[z]_0^2 \int_{0.5}^{2.5}\frac{1}{r}dr = 4\left[\ln\left[\frac{2.5}{.5}\right]\right] = 4\ln[5]$$

$$= 4 \times 1.609 = \boxed{6.436\, W_b}$$

$$\boxed{\psi = 6.436\, W_b}$$

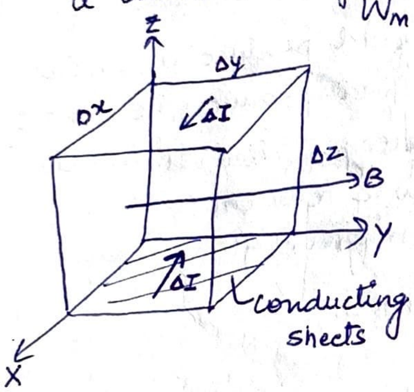
Energy stored in a Magnetic Field (Sessional 2019)

The potential energy stored in a electrostatic field was derived as

$$W_E = \frac{1}{2}\int \vec{D}\cdot\vec{E}\,dv = \frac{1}{2}\int \epsilon E^2 dV$$

The energy stored in a magnetic field is equal to the work needed to produce a current through the inductor.

$$W_m = \frac{1}{2}LI^2$$



Let the volume be covered with conducting sheets at the top and bottom surfaces with current ΔI.

$$\psi = \int \vec{B}\cdot\vec{ds}$$
$$\psi = \int \mu H\cdot\vec{ds}$$

$$\Delta L = \frac{\Delta\psi}{\Delta I} = \frac{\mu H\,\Delta x\Delta z}{\Delta I} \quad -(1)$$

where $\Delta I = H\,\Delta y$

$$\Delta W_m = \frac{1}{2}\Delta L\,\Delta I^2 = \frac{1}{2}\mu H^2 \Delta x\Delta y\Delta z$$

or $\Delta W_m = \frac{1}{2}\mu H^2 \Delta v$

The magnetostatic energy density w_m (in J/m^3) is defined as

$$w_m = \lim_{\Delta v\to 0}\frac{\Delta W_m}{\Delta v} = \frac{1}{2}\mu H^2 = \frac{1}{2}\vec{B}\cdot\vec{H}$$

Energy in a magnetostatic field in a linear medium is

$$W_m = \int w_m\,dv$$

or
$$\boxed{W_m = \frac{1}{2}\int \vec{B}\cdot\vec{H}\,dv = \frac{1}{2}\int \mu H^2 dv}$$

Permeability: It is a measure of the resistance of a material against the formation of a magnetic field i.e. it is a degree of magnetization that a material obtains in response to an applied magnetic field.

units → Henries/m
→ N/A^2

$$\mu_0 = 4\pi\times10^{-7}\ H/m = 12.57\times10^{-7}\ N/A^2$$

→ It is a measure of the amount of resistance encountered when forming a magnetic field in a vacuum.

Ampere's Law of Force

Ampere's force law states that the force of attraction or repulsion b/w two wires carrying currents is proportional to their lengths and the intensity of current passing through them. If current flows through them in the same direction, repulsion takes place. If current are flowing in opposite direction attraction takes place. (5)

The law is based on two basic concepts

1. Biot-savart law: which states that every current carrying wire produces a magnetic field around it.
2. Lorentz force which refers to the force that every magnetic field exerts on any electric charge moving in this field.

$$F = \underbrace{qE}_{\text{contribution of } \vec{E} \text{ Electric field}} + \underbrace{qvB}_{\text{contribution of magnetic field}}$$

$$F = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{L_1} \oint_{L_2} \frac{dI_1 \times dI_2 \times a_{r_{21}}}{R_{21}^2}$$

Magnetic Scalar and Vector Potential

Magnetic potential could be scalar (V_m) or vector potential (A).

Just like $E = -\nabla V$, the magnetic scalar potential V_m (in A) is

$$\boxed{H = -\nabla V_m} \quad \text{if } J = 0$$

we know $J = \nabla \times H$ ← from Ampere's law / Maxwell 4th eq

$= \nabla \times (-\nabla V_m)$

$= 0$

$$\left[\begin{array}{ll} \nabla \times (\nabla V) = 0 & [CG = 0] \\ \nabla \cdot (\nabla \times A) = 0 & [DC = 0] \end{array}\right]$$

Since V_m must satisfy the condition $\nabla \times (\nabla V)$ the magnetic scalar potential V_m is only defined in a region where $J = 0$

V_m satisfies Laplace eqn i.e. $\nabla^2 V_m = 0$ for ($J = 0$)

In magnetostatic field $\nabla \cdot \vec{B} = 0$ and to satisfy $\nabla \cdot (\nabla \times A)$ simultaneously we define vector magnetic vector potential A as

$$\boxed{B = \nabla \times A}$$

where $A = \int \frac{\mu_0 I dl}{4\pi R}$ (line current) or $A = \int \frac{\mu_0 K ds}{4\pi R}$ (Surface current) or $A = \int \frac{\mu_0 J dV}{4\pi R}$ (Volume current)

$$\psi = \int_S \vec{B} \cdot \vec{ds} = \underbrace{\int_S (\nabla \times \vec{A}) \cdot \vec{ds} = \oint \vec{A} \cdot \vec{dl}}_{\text{strokes theorem.}} \qquad \boxed{\psi = \oint \vec{A} \cdot \vec{dl}}$$

Que:- ~~If~~ ~~A~~

6)

Q:- If $A = 10r\,\hat{a}_z$ Wb/m (in cylindrical coordinate) in free space. Find the strength of the magnetic field. (sessional 2019)

Sol^n:
$$B = \nabla \times A = \frac{1}{r}\begin{vmatrix} \hat{a}_r & r\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial r} & \partial/\partial\phi & \frac{\partial}{\partial z} \\ A_r & rA_\phi & A_z \end{vmatrix} = \begin{vmatrix} \hat{a}_r & r\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 0 & 0 & 10r \end{vmatrix}$$

$$B = \frac{1}{r}\left[\hat{a}_r\left[\frac{\partial}{\partial\phi}10r - 0\right] - r\hat{a}_\phi\left[\frac{\partial}{\partial r}10r - 0\right] + \hat{a}_z[0-0]\right]$$

$$B = \frac{1}{r}\left[0 - r[10]\hat{a}_\phi + 0\right]$$

$$\boxed{B = -10\,\hat{a}_\phi}$$

Que:- Given magnetic vector potential $A = 10\sin\theta\,\hat{a}_\theta$ in spherical coordinate. Find flux density at $(2, \pi/2, 0)$ (ET, 2019)

Sol^n
$$B = \nabla\times A = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\,\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\,\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ 0 & r10\sin\theta & 0 \end{vmatrix}$$

$$= \frac{1}{r^2\sin\theta}\left[\hat{a}_r\left[\frac{\partial}{\partial\theta}\cdot 0 - \frac{\partial}{\partial\phi}(10\sin\theta)\right] - r\hat{a}_\theta\left[\frac{\partial}{\partial\phi}(0) - \frac{\partial}{\partial r}(0)\right] + \frac{\partial}{\partial\phi}\left[\frac{\partial}{\partial r}(r10\sin\theta) - \frac{\partial}{\partial\theta}(0)\right]r\sin\theta\,\hat{a}_\phi\right]$$

$$B = \frac{1}{r^2\sin\theta}\left[10\sin\theta\right]\hat{a}_\phi\cdot r\sin\theta = \frac{10\sin\theta}{r} = \frac{10\sin\frac{\pi}{2}}{2} = 5$$

~~If Another A = 10 sinθ aφ~~

~~B = ∇×A = ...~~

Que:- A current distribution gives rise to the vector magnetic potential $A = x^2y\,\hat{a}_x + y^2x\,\hat{a}_y - 4xyz\,\hat{a}_z$ Wb/m. Calculate

a) B at $(-1, 2, 5)$

b) The flux through the surface defined by $z = 1,\ 0 \le x \le 1,\ -1 \le y \le 4$

Sol^n:
$$B = \nabla\times A = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y & y^2x & -4xyz \end{vmatrix} = \left[\frac{\partial}{\partial y}(-4xyz) - \frac{\partial}{\partial z}y^2x\right]\hat{a}_x - \left[\frac{\partial}{\partial x}(-4xyz) - \frac{\partial}{\partial z}x^2y\right]\hat{a}_y + \left[\frac{\partial}{\partial x}xy^2 - \frac{\partial}{\partial y}x^2y\right]\hat{a}_z$$

$$= -4xz\,\hat{a}_x + 4yz\,\hat{a}_y + [y^2 - x^2]\,\hat{a}_z$$

At $(-1,2,5)$ $= -4(-1)(5)\hat{a}_x + 4(2)(5)\hat{a}_y + [2^2 - 1]\hat{a}_z$

$$\boxed{B = 20\hat{a}_x + 40\hat{a}_y + 3\hat{a}_z}$$

$$\psi = \int \vec{B}\cdot d\vec{s} = \int_{x=0}^{1}\int_{y=-1}^{4}(y^2 - x^2)\,dx\,dy\ \hat{a}_z\cdot\hat{a}_z \qquad ds = dx\,dy\,\hat{a}_z$$

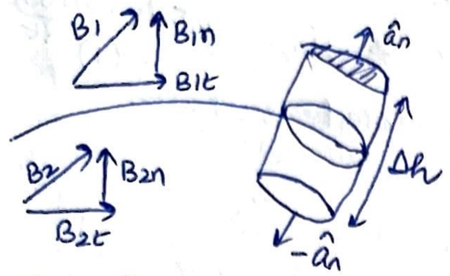
$$= \int_{y=-1}^{4}\left[y^2x - \frac{x^3}{3}\right]_0^1 dy = \int_{y=-1}^{4}\left(y^2 - \frac{1}{3}\right)dy = \left[\frac{y^3}{3} - \frac{1}{3}\right]_{-1}^{4} = \left[\frac{64}{3} + \frac{1}{3}\right] - \left[\frac{4}{3} + \frac{1}{3}\right] = \frac{65}{3} - \frac{5}{3} = \frac{60}{3}$$

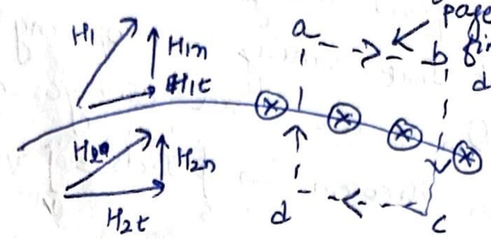
$$\boxed{\psi = 20\,W_b}$$

Boundary conditions in Magnetic Field [ET 2017, 2018, 2019 (6M)]

Gauss law of magnetic field and ampere's circuital law is used for determining the boundary condition.

i.e. $\oint \vec{B}\cdot\vec{ds} = 0$ and $\oint_L \vec{H}\cdot\vec{dl} = I$





$\oint \vec{B}\cdot\vec{ds} = 0$

$\underset{\text{Top}}{\oint \vec{B}\cdot\vec{ds}} + \underset{\text{surface}}{\cancel{\oint \vec{B}\cdot\vec{ds}}} + \underset{\text{bottom}}{\oint \vec{B}\cdot\vec{ds}} = 0$

$B_{1n}\Delta S - B_{2n}\Delta S = 0$

$\boxed{B_{1n} = B_{2n}}$ Normal component of $\vec{B}$ is continuous at the boundary

$B = \mu H$

$\mu_1 H_{1n} = \mu_2 H_{2n}$

$\boxed{\frac{H_{1n}}{H_{2n}} = \frac{\mu_2}{\mu_1}}$ Normal component of $\vec{H}$ is discontinuous at the boundary

$H_{1t}\Delta w - H_{1n}\frac{\Delta h}{2} - H_{2n}\frac{\Delta h}{2} - H_{2t}\Delta w + H_{2n}\frac{\Delta h}{2} + H_{1n}\frac{\Delta h}{2} = K\cdot\Delta w$ $\left(\because K = \frac{I}{b}\right)$

$\therefore (H_{1t} - H_{2t})\Delta w = K\Delta w$

$\boxed{H_{1t} - H_{2t} = K}$

→ Depends on surface current density K

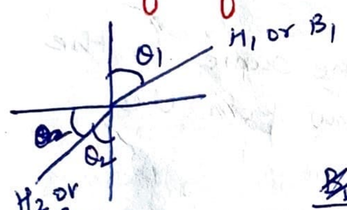
Tangential component of $\vec{H}$ is also discontinuous

If media is not conductor i.e. $\vec{K} = 0$

$H_{1t} = H_{2t}$

or $\boxed{\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}}$ → Discontinuous B at boundary condition

Law of Refraction of Magnetic Flux Lines



$B_{1n} = B_{2n}$ and $H_{1t} = H_{2t}$

$B_1\cos\theta_1 = B_2\cos\theta_2$

$H_1\sin\theta_1 = H_2\sin\theta_2$

$\frac{B_1}{\mu_1}\sin\theta_1 = \frac{B_2}{\mu_2}\sin\theta_2$

$\frac{\frac{B_1\sin\theta_1}{\mu_1}}{B_1\cos\theta_1} = \frac{\frac{B_2\sin\theta_2}{\mu_2}}{B_2\cos\theta_2}$ or $\boxed{\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2}}$

Que: There exists a boundary b/w two magnetic medium at $z=0$, $\mu = 4\mu_0$ H/m in region 1 $(z>0)$ and $\mu_2 = 7\mu_0$ in region 2 at $z<0$. If the magnetic flux density in the region 1 is $B_1 = 2\hat{a}_x - 3\hat{a}_y - 2\hat{a}_z$. Find B_2 and H_2 in region 2. There is no current at the boundary [ET 2017, 6.5 M]

Sol^n: $B_1 = 2\hat{a}_x - 3\hat{a}_y - 2\hat{a}_z$

$H_{1t} - H_{2t} = K$ $K = 0$ (given)

Since boundary is defined at $z=0$

$H_{1t} = H_{2t}$

$\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}$

$B_{1n} = -2\hat{a}_z = B_{2n}$

$B_{1t} = 2\hat{a}_x - 3\hat{a}_y$

$B_{2t} = \frac{\mu_2}{\mu_1}(\vec{B}_{1t}) = \frac{7}{4}(2\hat{a}_x - 3\hat{a}_y) = \boxed{3.5\hat{a}_x - 5.25\hat{a}_y}$

$\boxed{\vec{B}_2 = 3.5\hat{a}_x - 5.25\hat{a}_y - 2\hat{a}_z}$

$\vec{H}_2 = \frac{\vec{B}_2}{\mu_2} = \frac{1}{7\mu_0}(3.5\hat{a}_x - 5.25\hat{a}_y - 2\hat{a}_z) = \frac{1}{7\times4\pi\times10^{-7}}(3.5, -5.25, -2) = \frac{1}{0.879}(3.5, -5.25, -2)\times10^{-9}$

$\boxed{\vec{H}_2 = 3.98\hat{a}_x - 5.97\hat{a}_y - 2.27\hat{a}_z}$

8

Continuity Equation ~~and Relaxation Time~~ (Sessional 2019, 2020)

From the principle of charge conservation, the given rate of decrease of charge within a given volume must be equal to the net outward current flow through the surface of the volume.

Current I_{out} coming out of any closed surface is $I_{out} = \oint \vec{J} \cdot d\vec{s} = -\frac{dQ_{in}}{dt}$ — (1)

where Q_{in} is the total charge enclosed by the closed surface.

Divergence theorem $\rightarrow \oint_S \vec{J} \cdot d\vec{s} = \int_V \nabla \cdot \vec{J}\, dv$ — (2)

$$-\frac{dQ_{in}}{dt} = -\frac{d}{dt}\int \rho_v\, dv = -\int \frac{d\rho_v}{dt}\, dv \quad \text{— (3)}$$

Put (2) & (3) in (1)

$$\int_V \nabla \cdot \vec{J}\, dv = -\int \frac{\partial \rho_v}{\partial t}\, dv \quad \text{or} \quad \boxed{\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}} \quad \text{— (4)}$$

which is called the continuity of current equation or continuity equation or time varying equation of continuity.

Continuity equation is derived from the principle of conservation of charge and essentially states that there can be no accumulation of charge at any point.

For steady current $\frac{\partial \rho_v}{\partial t} = 0$ and hence $\nabla \cdot \vec{J} = 0$

$$\boxed{\nabla \cdot \vec{J} = 0} \quad \text{— (5)}$$

showing that the total charge leaving a volume is the same as the total charge entering it. Kirchoff's current law follow from this.

i.e. $\vec{J} = \sigma \vec{E}$ — (6)

and gauss law $\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon}$ — (7)

Put (6) and (7) in (4)

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$$

$$\nabla \cdot \sigma \vec{E} = -\frac{\partial \rho_v}{\partial t} = \cancel{\frac{\rho_v}{\epsilon}} \rightarrow \sigma \nabla \cdot \vec{E} = -\frac{\partial \rho_v}{\partial t} \rightarrow \sigma \frac{\rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$$

or $\frac{\partial \rho_v}{\partial t} + \frac{\sigma \rho_v}{\epsilon} = 0$

$$\frac{\partial \rho_v}{\rho_v} = -\frac{\sigma}{\epsilon}\, dt$$

Integrating both sides

$$\ln \rho_v = -\frac{\sigma t}{\epsilon} + \boxed{\ln \rho_{v_0}} \leftarrow \text{constant of integration}$$

$\boxed{\rho_v = \rho_{v_0} e^{-t/T_r}}$ where $T_r = \frac{\epsilon}{\sigma}$ where T_r is time constant in seconds

and ρ_{v_0} is initial charge density

The equation shows that the introduction of charge at some interior point of the material results in decay of volume charge density ρ_v.

The Time constant T_r is known as the relaxation time or rearrangment time.
Relaxation time is the time it takes a charge placed in the interior of a material to drop at e^{-1} (= 36.8%) of its initial value.

Inconsistency of Ampere Law / Displacement Current (June 2017, 2.5M)

For static EM field, we know $\nabla \times \vec{H} = \vec{J}$ — ①
Divergence of curl of any vector is zero i.e. DC = 0

$$\nabla \cdot (\nabla \times H) = 0 = \nabla \cdot J \quad — ②$$

The continuity equation $\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon}\rho_v = 0$, $\nabla J = -\frac{\partial \rho_v}{\partial t}$
requires $\nabla \cdot J = -\frac{\partial \rho_v}{\partial t} \neq 0$ — ③
However eq^n ① and ③ are obviously incompatible for time varying conditions. So, we need to modify eq ① to agree to eq ③. So, add a term in eq ① so that it becomes

$$\nabla \times H = J + J_d$$

where J_d is to be determined and defined

$$\nabla \cdot (\nabla \times H) = 0 = \nabla \cdot J + \nabla \cdot J_d \quad — ④$$

For ③ and ④ to agree

$$\nabla J_d = -\nabla J = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot D) = \nabla \frac{\partial D}{\partial t}$$

$$\boxed{J_d = \frac{\partial D}{\partial t}}$$

Substituting J_d in eq ①

$$\boxed{\nabla \times \vec{H} = J + \frac{\partial D}{\partial t}} \leftarrow \text{Ampere law with Maxwell equation}$$

This is Maxwell equation for a time-varying field. The term $J_d = \frac{\partial D}{\partial t}$ is known as displacement current density and J is the conduction current density ($J = \sigma E$)
Without the term J_d, the propogation of EM waves would be impossible. At low frequencies, J_d is usually neglected compared with J. However, at radio frequencies, the two terms are comparable.
Magnetic field can be generated in two ways
- → By electric current → original Ampere's law
- → By changing electric field → Maxwell addition

Magnetic field induced around any closed loop is proportional to the electric current plus displacement current (proportional to the rate of change of electric flux) through the enclosed surface.

Importance of Maxwell's addition to Ampere's Law

→ It makes the set of equation mathematically consistent for non-static field without changing law of Ampere & Gauss for static field.
→ It predicts that changing magnetic field produces electric field & vice verso.
These eq^n allow self-sustaining EM wave to Travel through empty space.

Que:- Find the displacement current density if $E = 6\times10^{-6}\sin(9\times10^{9}t)\, V/m$ (Sessional)

Soln: $E = 6\times10^{-6}\sin(9\times10^{9}t)\, V/m$

$$D = \epsilon E$$
$$= 8.854\times10^{-12}\times 6\times10^{-6}\sin(9\times10^{9}t)\, C/m^2$$
$$= 53.124\times10^{-18}\sin(9\times10^{9}t)\, C/m^2$$

$$J = \frac{\partial D}{\partial t} = 53.124\times10^{-8}\cos(9\times10^{9}t)(9\times10^{9})$$
$$\boxed{J = 478.124\times10^{-9}\cos(9\times10^{9}t)\, A/m^2}$$

Que: An AC voltage source $V = V_0 \sin wt$ is connected across a parallel plate capacitor C. Verify that the displacement current through the capacitor is same as the conduction current through the wire (6.5M, 2018)

Soln:
$$i = c\frac{dv}{dt} = \frac{C\partial V_0 \sin wt}{\partial t} = CV_0 w\cos wt$$

For a parallel plate capacitor of area A and plate seperation of

$$C = \frac{\epsilon_0 A}{d}$$

Electric field $\quad V = E.d \quad$ or $\quad E = \frac{V}{d}$

$$D = \epsilon E = \frac{\epsilon V_0 \sin wt}{d}$$

Displacement current $\rightarrow$ $i_d = \int \frac{\partial D}{\partial t}\cdot ds = \left(\frac{\epsilon A}{d}\right) V_0 w\cos wt$

$$\boxed{i_d = CV_0 w\cos wt = i}$$

Que:- Using Ampere circuital law in integral form. Find H if the volume current density distribution in cylindrical coordinate is given by $J = J_0\left(\frac{r}{a}\right)\hat{a}_z$ in the region $a < r < b$. Find magnetic vector potential A in the same region. (2018, 6.5M)

Soln
$a \le r < b$
$$\oint \vec{H}\cdot d\vec{l} = \int_S \vec{J}\cdot d\vec{s} = \int_{\phi=0}^{2\pi}\int_{r=a} J_0 \frac{r}{a}\hat{a}_z r\, dr\, d\phi\, \hat{a}_z$$

$ds = dr\cdot r d\phi \cdot \hat{a}_z$

$$\vec{H}\oint d\vec{l} = \int_{\phi=0}^{2\pi}\int_{r=a}^{b} J_0 \frac{r^2}{a}\, dr\, d\phi$$

$$\vec{H}\cdot 2\pi(r)_b = \frac{J_0}{a}\left[\frac{r^3}{3}\right]_a^b [2\pi]$$

$$\vec{H}\cdot 2\pi b = \frac{J_0 2\pi}{3a}(b^3 - a^3)\hat{a}_r$$

$$\vec{H} = \frac{J_0}{3ab}(b^3 - a^3)\hat{a}_r$$

Que: Find the value of displacement current density (J_d) and magnetic flux intensity (H) if $\vec{E} = 100\cos(100t - 20z)\,\hat{a}_y$ V/m. (Sessional 2020)

Sol^n:

$$E = 100\cos(100t - 20z)\,\hat{a}_y \text{ V/m}^2$$

$$D = \epsilon E$$

$$= 8.854\times10^{-12} \times 100\cos(100t - 20z)\,\hat{a}_y \text{ C/m}^2$$

$$= 8.854\times10^{-10}\cos(100t - 20z)\,\hat{a}_y$$

$$J_d = \frac{\partial D}{\partial t} = \frac{\partial}{\partial t}\left[8.854\times10^{-10}\cos(100t - 20z)\,\hat{a}_y\right]$$

$$= 8.854\times10^{-10}\left(-\sin(100t - 20z)\,\hat{a}_y\,(100)\right]$$

$$\boxed{J_d = -8.854\times10^{-8}\sin(100t - 20z)\,\hat{a}_y \text{ A/m}^2}$$

Faraday's law $\quad \nabla\times\vec{E} = -\mu\frac{\partial \vec{H}}{\partial t}$ or $\vec{H} = \frac{1}{\mu}\int(\nabla\times\vec{E})\,dt$

$$\nabla\times\vec{E} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y & 0 \end{vmatrix} = -\frac{\partial E_y}{\partial z}\,\hat{a}_x \text{ or}$$

$$= -2000\sin(100t - 20z)\,\hat{a}_x \text{ V/m.}$$

$$\vec{H} = -\frac{2000}{\mu}\int \sin(100t - 20z)\,dt\,\hat{a}_x \quad \text{V/m}$$

$$= \frac{2000}{\mu\times100}\cos(100t - 20z)\,\hat{a}_x \text{ V/m}$$

$$\boxed{\vec{H} = \frac{20}{\mu}\cos(100t - 20z)\,\hat{a}_x \text{ V/m}}$$

GATE Questions

Ques 1: Which of the following field equations indicate that the free magnetic charge do not exist.

a. $\vec{H} = \frac{1}{\mu}\nabla\times\vec{A}$ b. $\vec{H} = \oint \frac{I\vec{dl}\times\vec{R}}{4\pi R^2}$

c. $\nabla\cdot\vec{H} = 0$ d. $\nabla\times\vec{H} = \vec{J}$

Ans:- Magnetic monopoles or free charges do not exists are justified by $\nabla\cdot\vec{B} = 0$

$\nabla\cdot\mu\vec{H} = 0$

$\mu(\nabla\cdot\vec{H}) = 0$ $\boxed{\nabla\cdot\vec{H} = 0}$

Que: Match

A. $\nabla\times\vec{H} = \vec{J}$

B. $\oint\vec{E}\cdot\vec{dl} = -\oint_S \frac{\partial B}{\partial t}\cdot\vec{ds}$

C. $\nabla\cdot\vec{J} = -\frac{\partial\rho}{\partial t}$

(i) Continuity Equation

(ii) Faraday's law

(iii) Ampere's law

(iv) Gauss's Law

(v) Biot - Savart Law

Que:- The unit of $\nabla\times\vec{H}$ is

$\nabla\times\vec{H} = \vec{J}$ so $\frac{\text{Ampere}}{m^2}$.

Que:- If C is a closed curve enclosing a surface S, then the magnetic field intensity $\vec{H}$, the current density $\vec{J}$ and the electric flux density D are (2007)

$\rightarrow$ $\oint\vec{H}\cdot dl = \iint_S \left(\vec{J} + \frac{\partial D}{\partial t}\right)\cdot\vec{ds}$

Que: For static Electric and magnetic field in an inhomogeneous source-free medium, which of the following represents the correct form of two of the Maxwells equation. (2008)

(a) $\nabla\cdot\vec{E} = 0$, $\nabla\times\vec{B} = 0$

(b) $\nabla\cdot\vec{E} = 0$, $\nabla\cdot\vec{B} = 0$

(c) $\nabla\times\vec{E} = 0$, $\nabla\times\vec{B} = 0$

(d) $\nabla\times\vec{E} = 0$, $\nabla\cdot\vec{B} = 0$

Sol^n For any material, maxwell's eq^n are as follows

$\nabla\times\vec{E} = 0$ $\nabla\cdot\vec{B} = 0$

$\nabla\times\vec{H} = J$ $\nabla\cdot D = \rho_v$

For source free (charge or current free) conditions

$\nabla\times\vec{E} = 0$ $\nabla\cdot\vec{B} = 0$

$\nabla\times\vec{H} = 0$ $\nabla\cdot\vec{D} = 0$

$\nabla\times\frac{\vec{B}}{\mu} = 0$ $\nabla\cdot\epsilon\vec{E} = 0$

$\nabla\times\vec{B} = 0$ $\nabla\cdot\vec{E} = 0$

$$\boxed{\begin{matrix}\nabla\times\vec{B} = 0\\ \nabla\cdot\vec{E} = 0\end{matrix}}$$

Que: A current sheet $\vec{J} = 10\,\hat{a}_y$ A/m lies on the dielectric interface $x=0$ b/w two dielectric media with $\epsilon_{r1}=1$, $\mu_{r1}=1$ in Region-1 $(x<0)$ and $\epsilon_{r2}=2$, $\mu_{r2}=2$ in Region-2 $(x>0)$. If the magnetic field in Region-1 at $x=0^-$ is $\vec{H}_1 = 3\,\hat{a}_x + 30\,\hat{a}_y$. The $\vec{H}_2$ is (2011)

Sol^n:- Region is seperated by x

$$\vec{B}_{n1} = \vec{B}_{n2}$$

and $\vec{H}_{t1} - \vec{H}_{t2} = -\vec{J}_s \times \hat{a}_n$

$$\vec{B}_{n1} = \vec{B}_{n2}$$

$$\mu_1 \vec{H}_{x1} = \mu_2 \vec{H}_{x2}$$

$$\vec{H}_{x2} = \frac{\mu_1}{\mu_2} \times 3 = \frac{1}{2} \times 3 = 1.5\,\hat{a}_x$$

$$\vec{H}_{t1} - \vec{H}_{t2} = -10\,\hat{a}_y \cdot \hat{a}_x = +10\,\hat{a}_z$$

$$\vec{H}_{t2} = \vec{H}_{t1} - 10\,\hat{a}_z$$

$$\vec{H}_{t2} = 30\,\hat{a}_y - 10\,\hat{a}_z$$

$$\boxed{\vec{H}_2 = 1.5\,\hat{a}_x + 30\,\hat{a}_y - 10\,\hat{a}_z \text{ A/m}}$$